



LAST 10 YEARS NEET PYQ

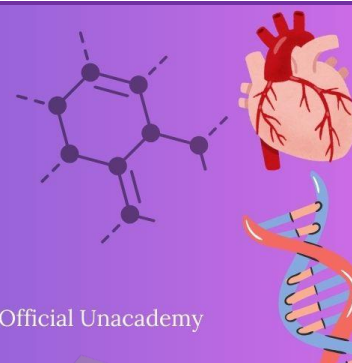
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Chapter 12 : Respiration in Plants

1. The complex II of mitochondrial electron transport chain is also known as [NEET-2025]

- (1) Cytochrome bc₁
- (2) Succinate dehydrogenase
- (3) Cytochrome c oxidase
- (4) NADH dehydrogenase

2. Match List-I with List-II: [Re-NEET-2024]

List-I

- A. ETS Complex I
- B. ETS Complex II
- C. ETS Complex III
- D. ETS Complex IV

List-II

- I. NADH Dehydrogenase
- II. Cytochrome bc₁
- III. Cytochrome C oxidase
- IV. Succinate Dehydrogenase

Choose the correct answer from the options given below :

- (1) A-IV, B-I, C-III, D-II
- (2) A-I, B-IV, C-II, D-III
- (3) A-III, B-I, C-IV, D-II
- (4) A-I, B-II, C-IV, D-III

3. Which of the following are correct about cellular respiration? [Re-NEET-2024]

- A. Cellular respiration is the breaking of C-C bonds of complex organic molecules by oxidation.
- B. The entire cellular respiration takes place in Mitochondria.
- C. Fermentation takes place under anaerobic condition in germinating seeds.
- D. The fate of pyruvate formed during glycolysis depends on the type of organism also.
- E. Water is formed during respiration as a result of O₂ accepting electrons and getting reduced.

Choose the correct answer from the options given below:

- (1) A, C, D, E only
- (2) A, B, E only
- (3) A, B, C, E only
- (4) B, C, D, E only

4. Match List I with List II

[NEET-2024]

List I

List II

A. Citric acid cycle

I. Cytoplasm

B. Glycolysis

II. Mitochondrial matrix

C. Electron transport system

III. Intermembrane space of mitochondria

D. Proton gradient

IV. Inner mitochondrial membrane

Choose the correct answer from the options given below:

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-III, B-IV, C-I, D-II

(4) A-IV, B-III, C-II, D-I

5. Identify the step in tricarboxylic acid cycle, which does not involve oxidation of substrate.

(1) Malic acid $\rightarrow$ Oxaloacetic acid

[NEET-2024]

(2) Succinic acid $\rightarrow$ Malic acid

(3) Succinyl-CoA $\rightarrow$ Succinic acid

(4) Isocitrate $\rightarrow$ α -ketoglutaric acid

6. Fatty acids are connected with the respiratory pathway through:

[NEET-2023-MANIPUR]

(1) Dihydroxy acetone phosphate

(2) Pyruvic acid

(3) Acetyl CoA

(4) α -Ketoglutaric acid

7. Match List I with List II :

[NEET-2023]

List I

List II

A. Oxidative decarboxylation

I. Citrate synthase

B. Glycolysis

II. Pyruvate dehydrogenase

C. Oxidative phosphorylation

III. Electron transport system

D. Tricarboxylic acid cycle

IV. EMP pathway

Choose the correct answer from the options given below :

(1) A - III, B - IV, C - II, D - I

(2) A - II, B - IV, C - I, D - III

(3) A - III, B - I, C - II, D - IV

(4) A - II, B - IV, C - III, D - I

8. Given below are two statements : One is labelled as Assertion A and other is labelled as Reason R :
Assertion A : ATP is used at two steps in glycolysis. [NEET-2023]

Reason R : First ATP is used in converting glucose into glucose-6-phosphate and second ATP is used in conversion of fructose-6-phosphate into fructose-1, 6-diphosphate.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is NOT the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

9. What is the net gain of ATP when each molecule of glucose is converted to two molecules of pyruvic acid ? [NEET-2022]

- (1) Six
- (2) Two
- (3) Eight
- (4) Four

10. What amount of energy is released from glucose during lactic acid fermentation? [NEET-2022]

- (1) More than 18%
- (2) About 10%
- (3) Less than 7%
- (4) Approximately 15%

11. The number of time(s) decarboxylation of isocitrate occurs during single TCA cycle is :

- (1) One [Re-NEET-2022]
- (2) Two
- (3) Three
- (4) Four

12. Which of the following statements is incorrect? [NEET-2021]

- (1) During aerobic respiration, role of oxygen is limited to the terminal stage.
- (2) In ETC (Electron Transport Chain), one molecule of $\text{NADH} + \text{H}^+$ gives rise to 2 ATP molecules, and one FADH_2 gives rise to 3 ATP molecules.
- (3) ATP is synthesized through complex V.
- (4) Oxidation-reduction reactions produce proton gradient in respiration

13. The number of substrate level phosphorylations in one turn of citric acid cycle is : [NEET-2020]

- (1) Three
- (2) Zero
- (3) One
- (4) Two

14. Pyruvate dehydrogenase activity during aerobic respiration requires :- [NEET-2020 COVID-19]

- (1) Calcium (2) Iron
- (3) Cobalt (4) Magnesium

15. Respiratory Quotient (RQ) value of tripalmitin is: [NEET-2019]

- (1) 0.9 (2) 0.7
- (3) 0.07 (4) 0.09

16. Conversion of glucose to glucose-6 phosphate, the first irreversible reaction of glycolysis, is catalyzed by: [NEET-2019]

- (1) Aldolase (2) Hexokinase
- (3) Enolase (4) Phosphofructokinase

17. Where is respiratory electron transport system (ETS) located in plants? [NEET-2019-ODISHA]

- (1) Mitochondrial matrix (2) Outer mitochondrial membrane
- (3) Inner mitochondrial membrane (4) Intermembrane space

18. What is the role of NAD⁺ in cellular respiration? [NEET-2018]

- (1) It functions as an enzyme
- (2) It functions as an electron carrier
- (3) It is a nucleotide source for ATP synthesis
- (4) It is the final electron acceptor for anaerobic respiration

19. Which of these statements is incorrect? [NEET-2018]

- (1) Enzymes of TCA cycle are present in mitochondrial matrix.
- (2) Glycolysis occurs in cytosol.
- (3) Glycolysis operates as long as it is supplied with NAD that can pick up hydrogen atoms.
- (4) Oxidative phosphorylation takes place in outer mitochondrial membrane.

20. Which statement is wrong for Krebs' cycle? [NEET-2017]

- (1) There is one point in the cycle where FAD⁺ is reduced to FADH₂
- (2) During conversion of succinyl CoA to succinic acid, a molecule of GTP is synthesised
- (3) The cycle starts with condensation of acetyl group (acetyl CoA) with pyruvic acid to yield citric acid
- (4) There are three points in the cycle where NAD⁺ is reduced to NADH + H⁺

21. Which of the following biomolecules is common to respiration-mediated breakdown of fats, carbohydrates and proteins? [NEET- II 2016]

- (1) Pyruvic acid (2) Acetyl CoA
- (3) Glucose-6-phosphate (4) Fructose 1,6-bisphosphate

ANSWER KEY

1. (2)
2. (2)
3. (1)
4. (2)
5. (3)
6. (3)
7. (4)
8. (1)
9. (2/3)
- 10.(3)
- 11.(2)
- 12.(2)
- 13.(3)
- 14.(4)
- 15.(2)
- 16.(2)
- 17.(3)
- 18.(2)
- 19.(4)
- 20.(3)
- 21.(2)